

Project Interim Report

On

Weather Monitoring System (Using Core Java & OOP Concepts)

RU-100-01-00012

Object Oriented Programming With Java

SESSION: 2025-26



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◆ Abstract

The Weather Monitoring System is a Java-based console application designed to record, analyze, and manage environmental data such as temperature and humidity. The system helps in monitoring weather conditions and generating alerts when predefined thresholds are exceeded, such as high temperature or high humidity.

This project is needed to demonstrate how real-world monitoring systems work in a simplified manner using programming concepts. Manual tracking of environmental data can be inefficient and error-prone, so this system provides an automated way to store and process weather information.

The application allows users to input weather data, store multiple records using arrays, search specific data, and save or load records from a file. It uses object-oriented programming concepts such as encapsulation, constructors, and methods. Additionally, it implements file handling and exception handling to ensure efficient and secure data processing. Overall, the system simulates a basic weather monitoring application and enhances understanding of Java programming concepts.

◆ Introduction

The Weather Monitoring System is a console-based application developed using Java and object-oriented programming principles. It is designed to monitor environmental conditions like temperature and humidity and generate alerts based on predefined conditions.

In real-world scenarios, weather monitoring plays an important role in agriculture, disaster management, and daily life. This project simulates such systems in a simplified way, helping users understand how data is collected, processed, and analyzed.

The application provides features such as adding weather records, displaying all stored data, searching records, and saving/loading data using file handling. It also includes a menu-driven interface, making it user-friendly.

This project mainly focuses on applying core Java concepts like classes, objects, arrays, conditional statements, and exception handling to solve a real-life problem efficiently.

◆ Objectives

- To monitor temperature and humidity data
- To generate alerts based on weather conditions
- To store and manage multiple weather records
- To reduce manual calculation and tracking
- To implement file handling for data storage
- To apply object-oriented programming concepts

◆ Scope

This project is suitable for small-scale applications such as basic environmental monitoring and educational purposes. It can be used in schools or small organizations to track simple weather data.

In the future, this system can be enhanced by integrating a database for large-scale data storage, adding a graphical user interface (GUI), and connecting with real-time weather APIs to fetch live data from the internet.

◆ System Requirements

Hardware Requirements:

- Computer/Laptop with minimum 4GB RAM
- Processor: Intel i3 or above

Software Requirements:

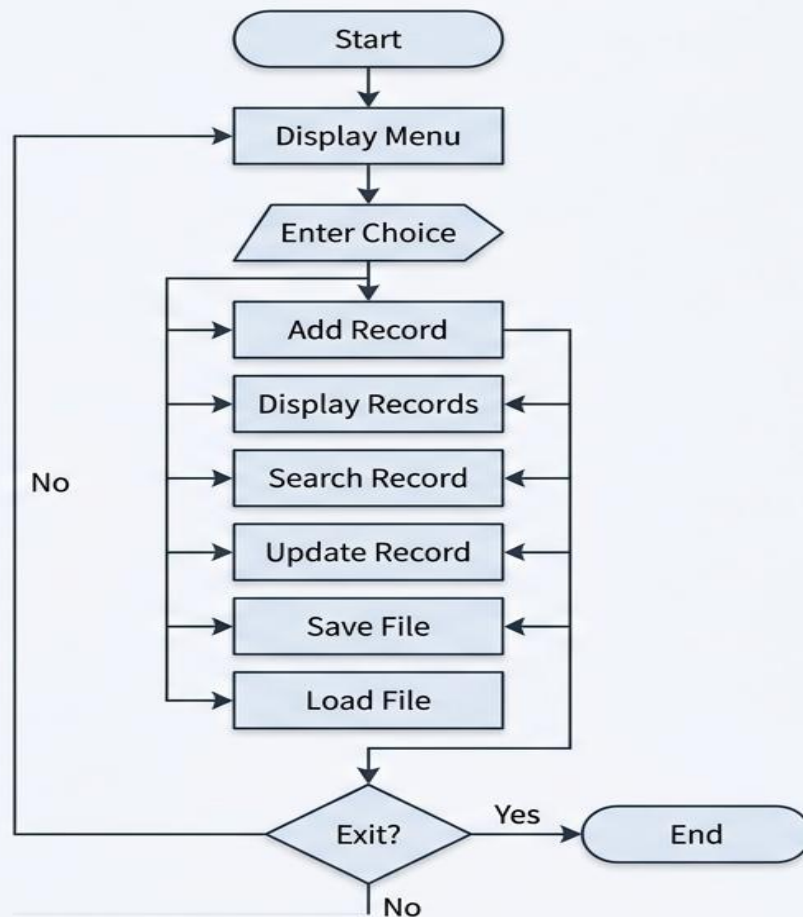
- Java JDK (Java Development Kit)
- IDE: VS Code / Eclipse / IntelliJ IDEA
- Operating System: Windows/Linux/Mac

◆ Methodology

The working of the system follows a step-by-step approach:

1. The program starts and displays a menu to the user.
2. The user selects an option such as adding a record, viewing records, or searching data.
3. When adding a record, the user inputs temperature and humidity.
4. The system creates an object of the WeatherData class and stores it in an array.
5. The alert is automatically generated using conditional logic.
6. The user can view all records or search for specific data.
7. Data can be saved to a file or loaded from a file using file handling.
8. The program continues until the user chooses to exit.

◆ Flowchart



◆ Classes-Methods/Variables Used

Class: WeatherData

Variables:

- temperature (double)
- humidity (double)
- alert (String)

Methods:

- Constructor WeatherData()
- getTemperature()
- getHumidity()
- getAlert()
- generateAlert()
- display()
- toFileString()

Class: WeatherApp

Variables:

- Scanner sc
- WeatherData[] records (Array)

- int count
- Methods
- addRecord()
- displayRecords()
- searchRecord()
- saveToFile()
- loadFromFile()
- menu()
- main()

◆ Implementation

The project consists of two main classes: WeatherData and WeatherApp.

The WeatherData class is responsible for storing temperature and humidity values. It uses encapsulation by keeping variables private and provides getter methods to access them. It also contains a method to generate alerts based on conditions.

The WeatherApp class is the main class that controls the program flow. It uses an array to store multiple weather records. A menu-driven approach is used to allow users to perform operations such as adding, displaying, searching, saving, and loading records.

File handling is implemented using FileWriter and Scanner classes to store and retrieve data from a file. Exception handling is used to handle invalid inputs and file errors.

◆ Source-code-Weather-Monitoring App

◆ Sample Input / Output (Write this in report)

```

PS C:\BTECH G1\JAVA-Project> cd "c:\BTECH G1\JAVA-Project\" ; if ($?) { javac WeatherApp.java } ; if ($?) { java WeatherApp }

===== Weather Monitoring System =====
1. Add Weather Record
2. Display All Records
3. Search Record
4. Save to File
5. Load from File
6. Exit
Enter choice: 1
Enter Temperature: 42
Enter Humidity: 85
Record Added Successfully!

```

◆ Gantt Chart

Task	Duration
Planning & Design	1 Day
Coding (Classes & Logic)	2 Days
Testing	1 Day

File Handling Integration	1 Day
Documentation	1 Day
Total Estimated Time	6 Days

◆ Conclusion

The Weather Monitoring System project successfully demonstrates the use of Java and object oriented programming concepts in solving a real-world problem. It provides a functional system to store, analyze, and manage weather data efficiently.

The project helped in understanding key concepts such as encapsulation, arrays, file handling, and exception handling. It also improved logical thinking and problem-solving skills. Overall, this system serves as a strong foundation for developing more advanced applications in the future.